MILITARY STANDARD

MIL-STD-1320-209

(NAVY)

23 MARCH 1982

TRUCKLOADING

STANDARD MISSILE (MR)

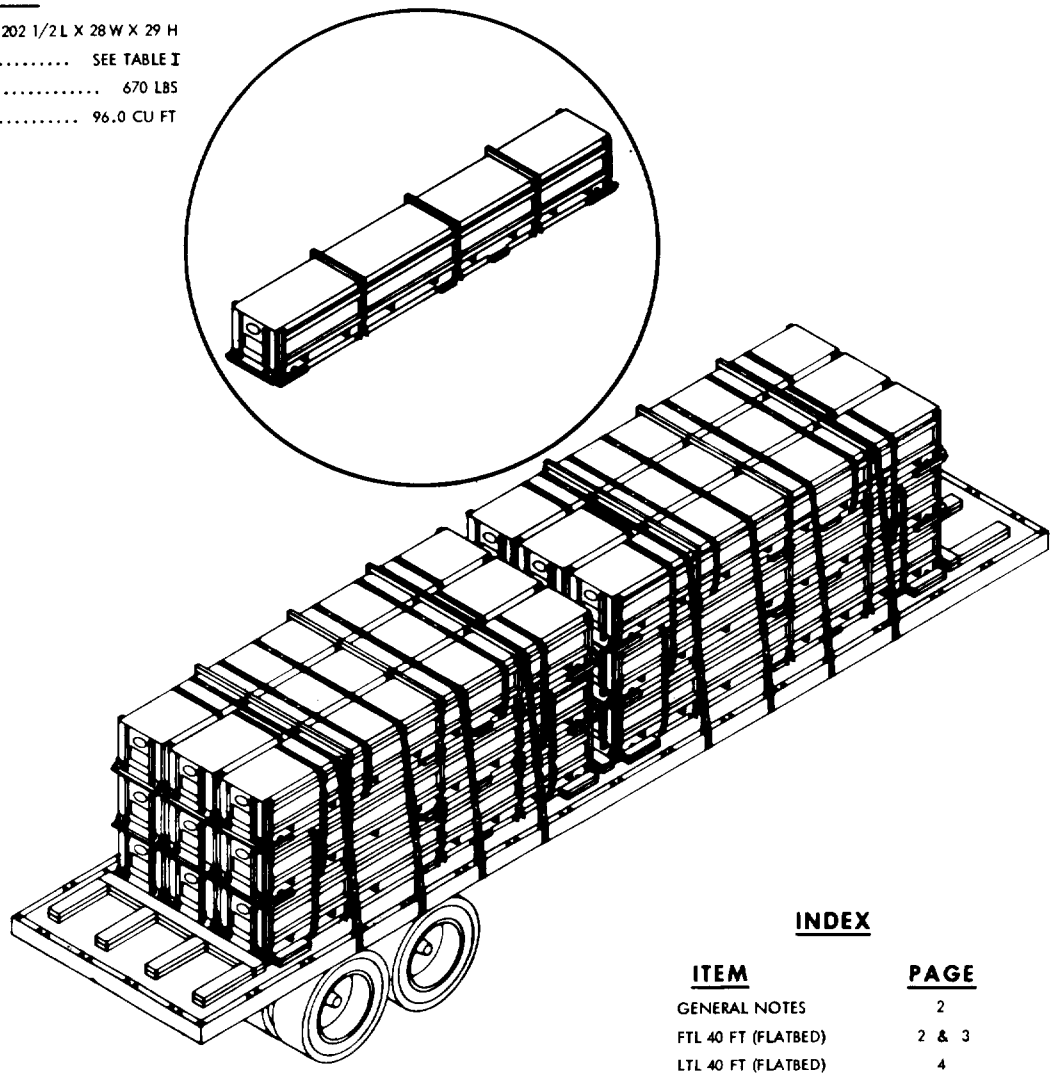
RIM-66A, TYPE 1; RIM-66C; RGM-66D-1, 2 OR

ROCKET MOTOR MK 56 & MODS (DTRM)

IN CONTAINER MK 372 MODS 2, 3 OR 5

CONTAINER DATA

DIMENSIONS ..... 202 1/2 L X 28 W X 29 H
GROSS WEIGHT ..... SEE TABLE I
TARE WEIGHT ..... 670 LBS
CUBE ..... 96.0 CU FT



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NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

FSC 8140

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M S Gray NWHC 3/17/82
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J E Kelly 3/17/82
SIGNATURE SEASYSCOM BY DIRECTION DATE

ORIGINATOR 3/16/82
DATE

NAVAL WEAPONS HANDLING CENTER
WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE STANDARD MISSILE RIM-66A TYPE 1, RIM-66C OR RGM-66D-1, 2 AND THE DUAL THRUST ROCKET MOTOR (DTRM) MK 56 AND MODS IN THE CONTAINER MK 372 MODS 2, 3 OR 5.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT FLATBED TRAILERS WITH THE TRAILERS TANDEM AXLES LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER). DO NOT USE TRAILERS WITH METAL FLOORS.
3. WHEN THE CONTAINERS ARE TWO HIGH, THEY SHALL BE UNITIZED AS SHOWN ON PAGE 6.

WARNING

TO AVOID TOPPLING SPECIAL CARE SHOULD BE TAKEN WHEN MOVING UNITIZED CONTAINERS. THE UNITIZING OF THE CONTAINERS IS AUTHORIZED AS PART OF THE TRUCKLOADING PROCEDURE ONLY. UNITIZING IS TO ENSURE CONTINUOUS ENGAGEMENT OF THE STACKING FEATURES AND MOVEMENT OF UNITIZED CONTAINERS SHOULD BE LIMITED TO THAT NECESSARY TO LOAD OR UNLOAD THE TRAILER.

4. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
5. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO AN ACCESSIBLE AREA AND ATTACH APPROPRIATE EXPLOSIVES PLACARD AS SPECIFIED IN TABLE I TO BOTH SIDES, FRONT AND BACK OF THE TRAILER.
6. UNLESS OTHERWISE SPECIFIED, ALL MATERIAL SHALL BE AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320.
7. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320.

FTL 40 FT. TRAILER (FLATBED)

1. EACH STACK OF CONTAINERS SHALL BE SECURED TOGETHER WITH TWO 1 1/4 X .035 CROSS STRAPS. THE STRAPS PASS OVER THE COVERS OF THE CONTAINERS IN THE TOP LAYER AND RETURN PASSING UNDER THE BOTTOMS OF THESE CONTAINERS, BINDING THE TOP LAYER TOGETHER.
2. LOAD BINDERS AND CHAINS MAY BE USED AS AN ALTERNATE FOR THE 2 INCH STRAPPING SUBSTITUTE ONE CHAIN AND LOAD BINDER FOR EACH 2 INCH STRAP. LAMINATED DOUBLED 2 X 6'S SHALL BE USED OVER THE TOP OF THE CONTAINERS TO PREVENT DAMAGE BY THE CHAIN AND THE CHAIN SHALL BE SECURED TO THE BOARDS WITH FOUR 20d NAILS BY DRIVING THE NAIL THROUGH THE CHAIN LINK INTO THE BOARD AND CLINCHING THE NAIL OVER THE CHAIN.
3. CHAINS AND LOAD BINDERS SHALL MEET THE FOLLOWING CONDITIONS:
   a. CHAIN AND FITTINGS SHALL BE 3/8 INCH AND LOAD BINDERS 3/8 INCH CAPACITY.
   b. CHAIN, FITTINGS AND LOAD BINDERS SHALL HAVE A MINIMUM BREAK LOAD OF 9,800 LBS.
   c. LOAD BINDERS SHALL BE SAFETY WIRED WITH 16 GAUGE SOFT ANNEALED IRON WIRE OR SECURED USING THE END PIECE OF THE 3/8 CHAIN.

   PRIOR TO LOADING THE TRAILER AND DURING THE PRELOADING INSPECTION REQUIRED BY NAVWEPS OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS AND LOAD BINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS AND LOAD BINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOAD BINDER.

TABLE I

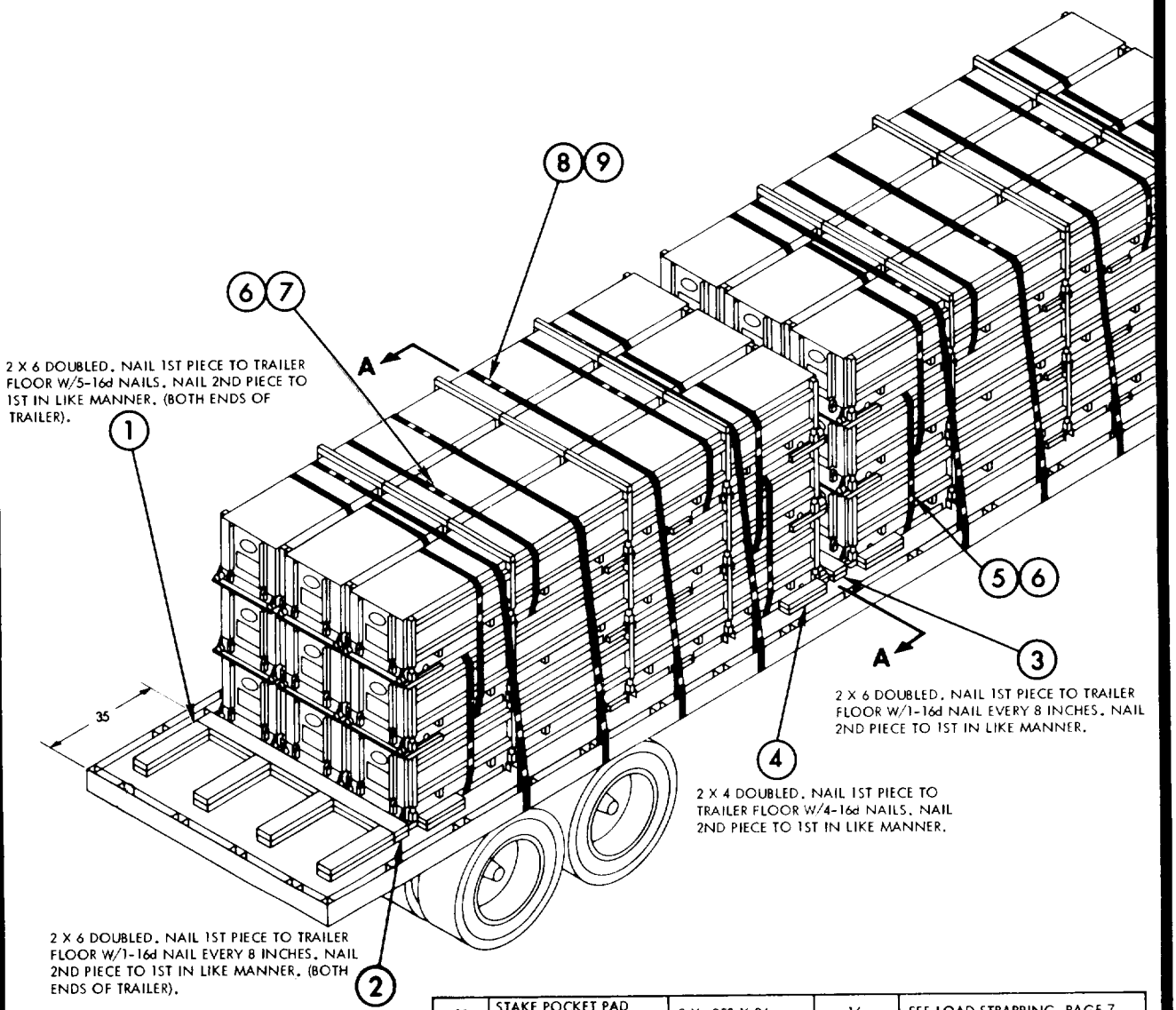
MISSILE OR DTRM	LOADED CONTAINER WT. (LBS.) APPROX.	FTL WEIGHT (LBS.) APPROX.	DOT HAZARD CLASSIFICATION
RIM-66A, TYPE I	2038	36,684	EXPLOSIVES A
RIM-66C	2090	37,620	EXPLOSIVES A
RGM-66D-1,2	2146	38,628	EXPLOSIVES A
DTRM MK 56	1585	28,530	EXPLOSIVES B

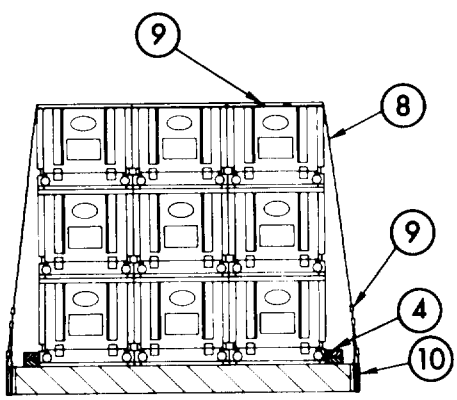
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TRUCKLOAD DATA

NUMBER OF UNITS 18

WEIGHT ..... SEE TABLE I





SECTION A-A

PIECE NO.	DESCRIPTION	SIZE	NO. PCS. REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
10	STAKE POCKET PAD (STRAPPING)	2 X .050 X 24	16	SEE LOAD STRAPPING, PAGE 7		
9	SEAL	FOR 2 IN. STRAP	64	-	-	-
8	TIE DOWN STRAP	2 X .050 X 29 FT.	8	SEE LOAD STRAPPING, PAGE 7		
7	CROSS STRAP	1 1/4 X .035 X 20 FT 8 IN.	4	-	-	-
6	SEAL	FOR 1 1/4 STRAP	56	-	-	-
5	UNITIZING STRAP	1 1/4 X .035 X 16 FT	24	SEE PAGE 6		
4	SLEEPER	2 X 4 X 24	16	SEE FIELD NOTE		
3	CROSS MEMBER	2 X 6 X 90	2	SEE FIELD NOTE		
2	CROSS MEMBER	2 X 6 X 90	4	SEE FIELD NOTE		
1	END CLEAT	2 X 6 X 28	16	SEE FIELD NOTE		

LIST OF MATERIAL & NAILING DATA

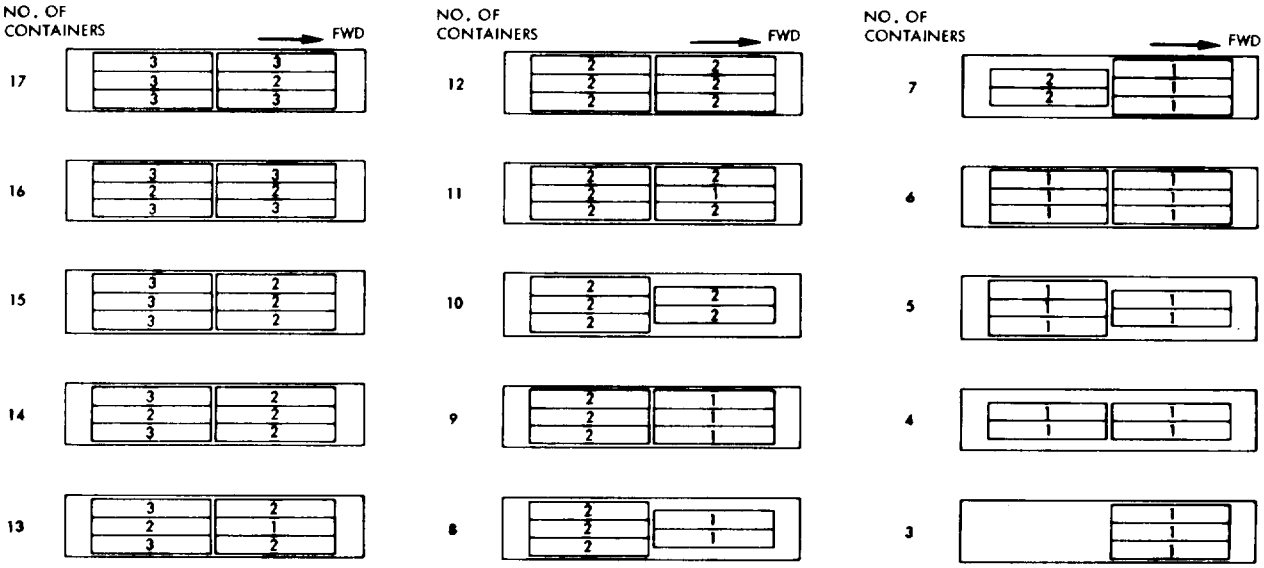
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LTL 40 FT. TRAILER (FLATBED)

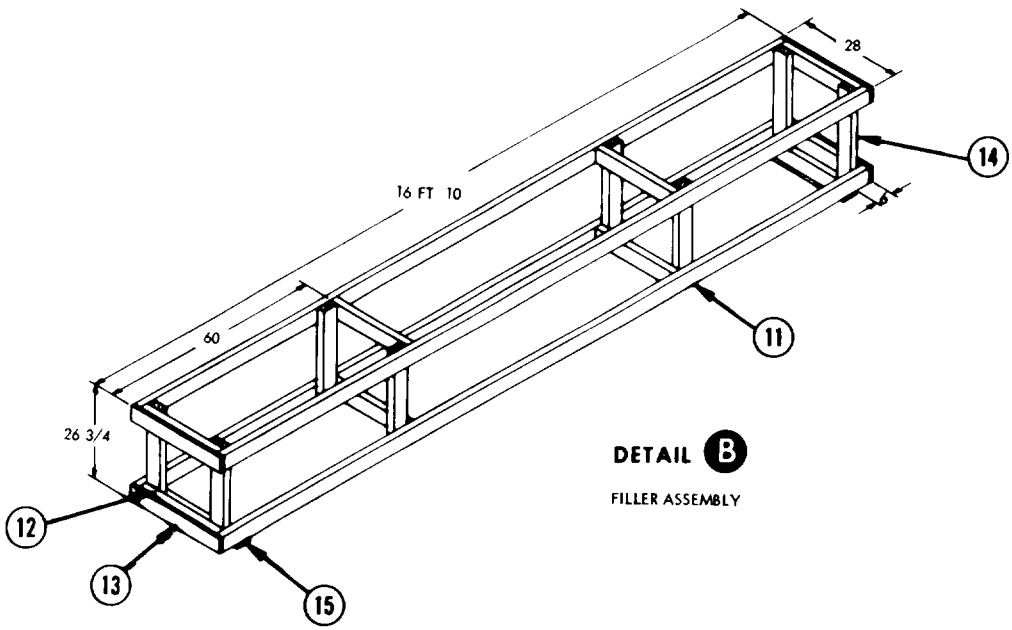
1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD, SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER OF CONTAINERS TO BE SHIPPED FROM TABLE II AND POSITION CONTAINERS ON THE TRAILER ACCORDINGLY.
2. WHEN THE CENTER OF THE TOP LAYER OF A STACK DOES NOT HAVE A CONTAINER (SEE LOAD PLAN FOR 14 CONTAINERS), THE FILLER ASSEMBLY, DETAIL B SHALL BE USED. SECURE THE ASSEMBLY TO THE TIE DOWNS TO PREVENT LONGITUDINAL MOVEMENT.
3. ALL STACKS OF THREE (3) HIGH CONTAINERS SHALL BE SECURED TO THE TRAILER WITH FOUR (4) STRAPS, (OR CHAINS). STACKS OF CONTAINERS TWO (2) HIGH SHALL BE SECURED WITH THREE (3) STRAPS, (OR CHAINS).
4. THE LTL SHOWN ON PAGE 5 IS FOR FOURTEEN (14) CONTAINERS AND SHOWS CORRECT BLOCKING, BRACING AND TIE DOWNS FOR A TRUCKLOAD OF THIS QUANTITY.
5. THE LTL PLAN SHOWN ON PAGE 5 DOES NOT LIMIT SHIPMENTS TO THIS QUANTITY. ITS PRESENTATION IS INTENDED TO EXPLAIN THE CORRECT METHODS OF LOADING LTL SHIPMENTS.

TABLE II

LOAD PATTERN *



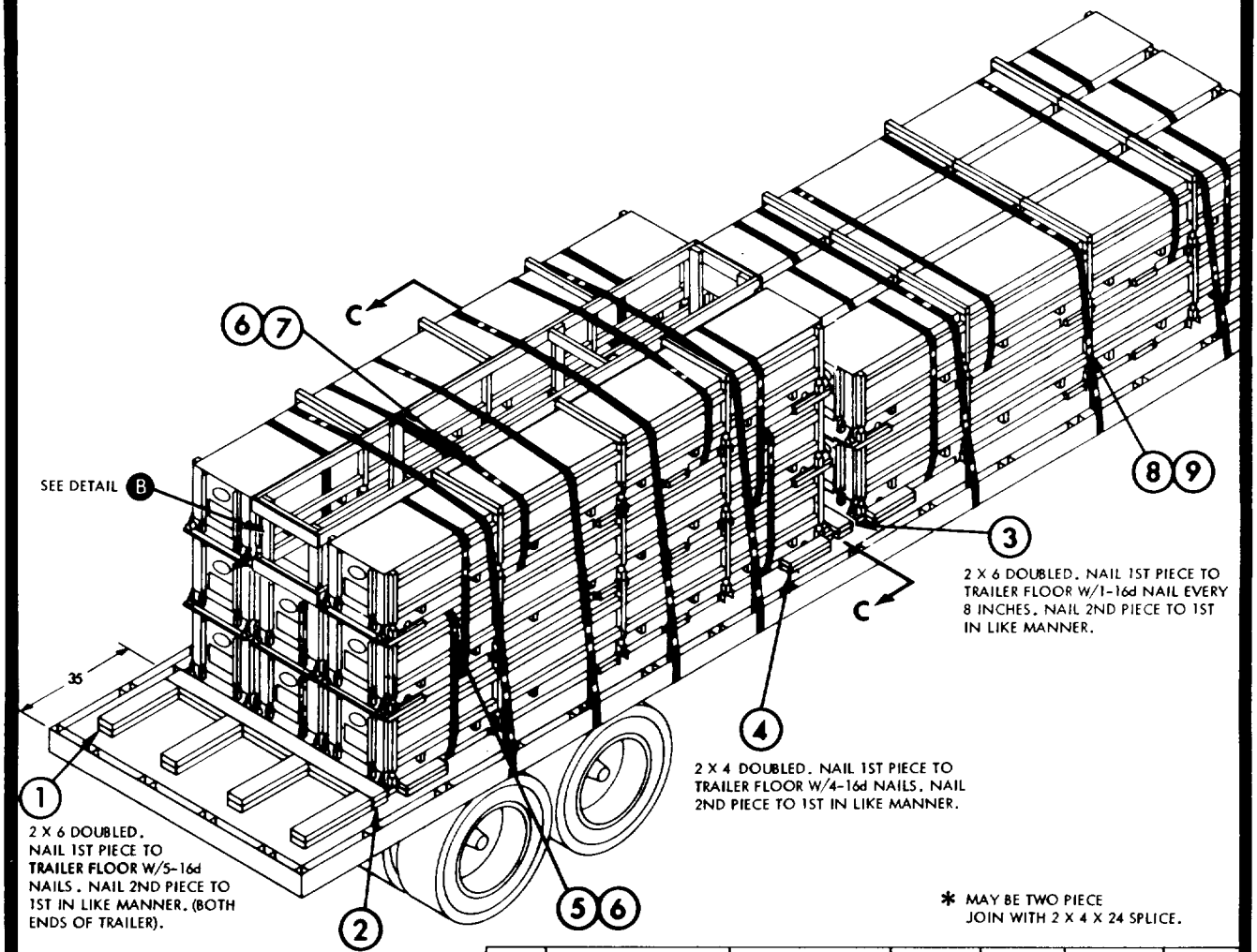
* LOAD PATTERN SHOWS FLOOR PLAN FOR TRAILER. NUMBER INDICATES THE NUMBER OF CONTAINERS IN A LAYER.



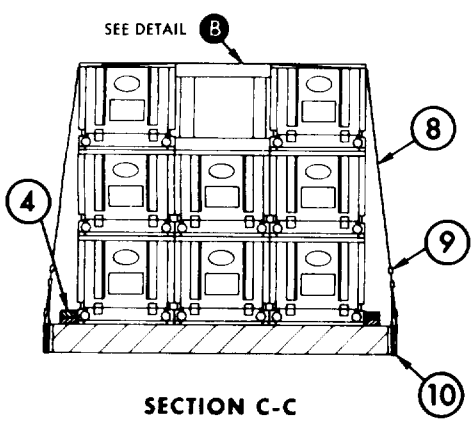
DETAIL B

FILLER ASSEMBLY

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* MAY BE TWO PIECE JOIN WITH 2 X 4 X 24 SPLICE.



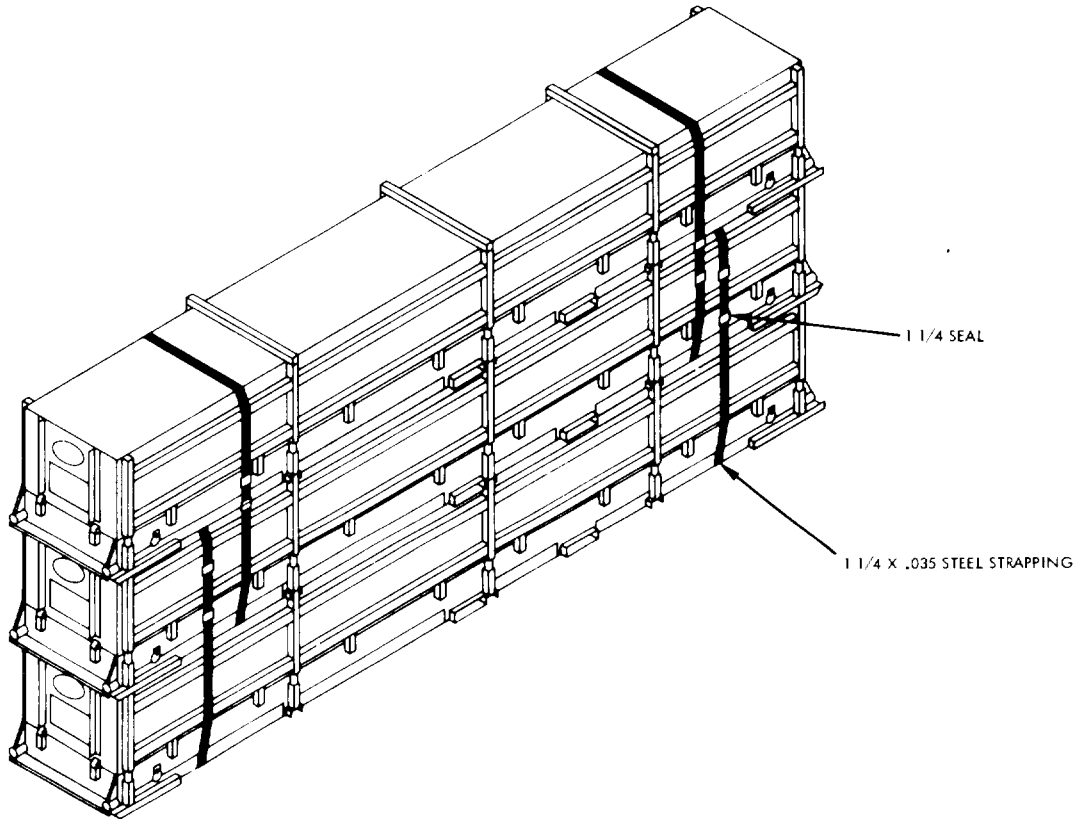
SECTION C-C

PIECE NO.	DESCRIPTION	SIZE	NO. PCS. REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
15	SUPPORT	1 X 4 X 28	2	11	2/JOINT	8d
14	VERTICAL	2 X 4 X 26 3/4	8	SEE 11 & 14	-	-
13	END PIECE	1 X 4 X 28	4	11 & 12	2/JOINT	6d
12	CROSS BRACE	2 X 4 X 25	8	14	2/JOINT	10d
11	HORIZONTAL PIECE	2X4X 16 FT -10 *	4	12 & 14	2/JOINT	16d
10	STAKE POCKET PAD (STRAPPING)	2 X .050 X 24	14	SEE LOAD STRAPPING, PAGE 7		
9	SEAL	FOR 2 IN. STRAP	56	-	-	-
8	TIE DOWN STRAP	2X .050 X 29 FT	7	SEE LOAD STRAPPING, PAGE 7		
7	CROSS STRAP	1 1/4 X .035 X 20 FT 8 IN.	4	-	-	-
6	SEAL	FOR 1 1/4 STRAP	40	-	-	-
5	UNITIZING STRAP	1 1/4 X .035 X 16 FT	16	SEE PAGE 6		
4	SLEEPER	2 X 4 X 24	16	SEE FIELD NOTE		
3	CROSS MEMBER	2 X 6 X 90	2	SEE FIELD NOTE		
2	CROSS MEMBER	2 X 6 X 90	4	SEE FIELD NOTE		
1	END CLEAT	2 X 6 X 28	16	SEE FIELD NOTE		

LIST OF MATERIAL & NAILING DATA

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UNITIZING AND HANDLING



CONTAINER UNITIZING AND BANDING PROCEDURE:

1. PLACE TWO OR MORE PIECES OF LUMBER CROSSWISE ON FLOOR OR GROUND NEXT TO MISSILE CONTAINER.
2. USING FORK LIFT TRUCK OR AN APPROPRIATE HOISTING DEVICE EQUIPPED WITH MK 77 MOD 3 HOISTING SLING, STACK THREE CONTAINERS ON THE LUMBER PIECES.
3. BE CERTAIN THAT NESTING FEATURES AND BEARING SURFACES ARE PROPERLY POSITIONED TO PREVENT SHIFTING IN LATERAL AND LONGITUDINAL DIRECTIONS.
4. SECURE BOTTOM CONTAINERS TO CENTER CONTAINER WITH TWO STRAPS POSITIONED AS INDICATED ABOVE.
5. SEAL EACH STRAP WITH TWO SEALS AND DOUBLE CRIMP.
6. REPEAT THIS OPERATION TO SECURE CENTER CONTAINERS TO TOP CONTAINER.

CONTAINER HANDLING PROCEDURE:

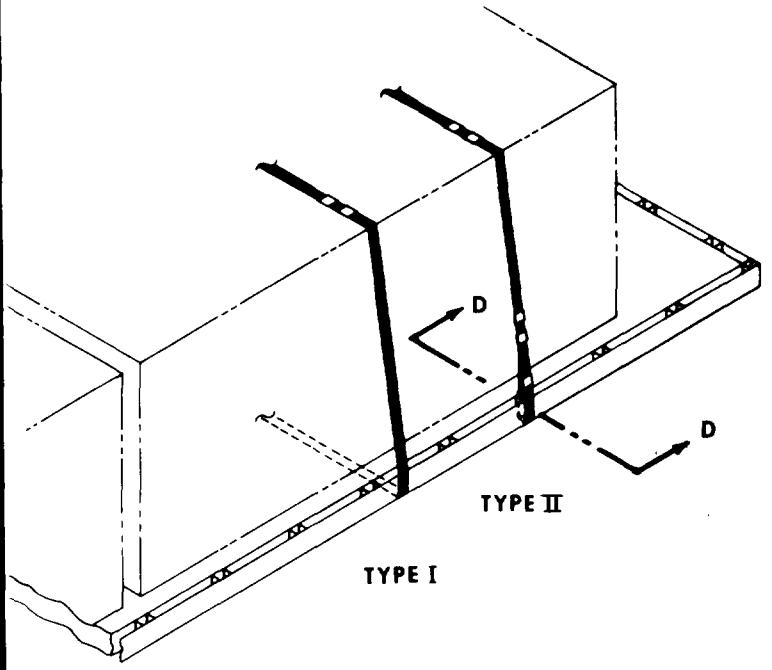
1. UNITIZED CONTAINER MK 372 MODS 2, 3 OR 5 FOR STANDARD MISSILE ARE HANDLED AND POSITIONED ON FLATBED TRAILERS WITH A SUITABLE FORK LIFT TRUCK. IF NECESSARY CONTAINERS CAN BE LOADED ONE AT A TIME USING AN APPROPRIATE HOISTING DEVICE EQUIPPED WITH MK 77 MOD 3 HOISTING SLING, AND UNITIZED IN POSITION.

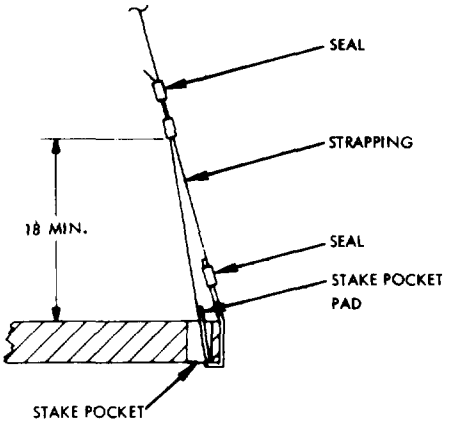
WARNING

TO AVOID TOPPLING, SPECIAL CARE SHOULD BE TAKEN WHEN MOVING STACKS OF CONTAINERS. THE UNITIZING OF THE CONTAINERS IS AUTHORIZED AS PART OF THE TRUCKLOADING PROCEDURE ONLY. UNITIZING IS TO ENSURE CONTINUOUS ENGAGEMENT OF THE STACKING FEATURES AND MOVEMENT OF UNITIZED CONTAINERS SHOULD BE LIMITED TO THAT NECESSARY TO LOAD AND UNLOAD THE FLATBED TRAILER. CONTAINERS SHALL BE DEUNITIZED AFTER UNLOADING THE VEHICLE.

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LOAD STRAPPING





SECTION D-D

THE FOLLOWING TWO METHODS OF TIE-DOWN STRAPPING ARE APPROVED FOR USE WITH THE SHIPMENT OF LOADS ON FLATBED TRAILERS. STEEL STRAPPING SHALL BE IN ACCORDANCE WITH FEDERAL SPECIFICATION QQ-S-781. STRAPPING SEALS SHALL HAVE A MINIMUM TENSILE STRENGTH EQUAL TO 75 PERCENT OF THE STRAP STRENGTH. FOR EACH OF THE TYPES ILLUSTRATED ABOVE IT IS PREFERRED TO POSITION, TENSION AND DOUBLE CRIMP THE STRAP SEALS AT THE TOP OF THE LOAD.

TYPE I - CONTINUOUS STRAPPING AROUND THE FLATBED AND THE LOAD.

TYPE II - THE STRAPPING IS SECURED TO THE STAKE POCKETS, ONE PIECE ON EACH SIDE OF THE TRAILER, AND IS BROUGHT UP OVER THE LOAD, TENSIONED, AND SEALED WITH THE DOUBLE-CRIMPED SEALS ON THE TOP. METHOD OF SECURING STRAPPING TO STAKE POCKET IS SHOWN IN SECTION D-D. THE SHORT END IS ON THE INSIDE AND IS SECURED WITH TWO DOUBLE-CRIMPED SEALS AT A MINIMUM OF 18 INCHES ABOVE THE TRAILER BED. A STAKE POCKET PAD (A SHORT PIECE OF THE SAME STRAPPING 18 INCHES LONG) IS INSERTED BETWEEN THE MAIN STRAP AND THE STAKE POCKET AND IS SECURED TO THE MAIN STRAP WITH A SEAL AS SHOWN.

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No. 22-R255

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MIL-STD-1320-209

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